

Design and Kinematic Verification of a Roller Chain and Sprocket Drive Assembly

Parametric Solid Modeling, Assembly Simulation, and Manufacturing Drawing Package

Prepared using PTC Creo Parametric | ME-430 Introduction to Computer-Aided Design | New Jersey Institute of Technology

1. Project Overview

This project covers the complete design cycle of a single-pitch segment of a mechanical roller chain and sprocket drive: parametric modeling of every component from first principles, bottom-up assembly of the chain in Creo Parametric, verification of correct kinematic fit between mating parts, and generation of a full manufacturing drawing package. The chain is one of the most common machine elements used to transmit rotational power between shafts, and it is a compact but demanding case study in assembly design because every joint in the chain must simultaneously satisfy running clearance for articulation, load transfer through the pins, and repeatability across dozens of identical links.

The final assembly is a five-part, twenty-instance system — roller link plates, pin link plates, bushings, rollers, and pins — built as two reusable sub-assemblies (a roller-link module and a pin-link module) that are mirrored and combined to form the complete chain segment shown in Figure 1. Building the chain from sub-assemblies rather than as twenty individually placed parts kept the model parametric and made it possible to propagate a single dimension change through every repeated instance.

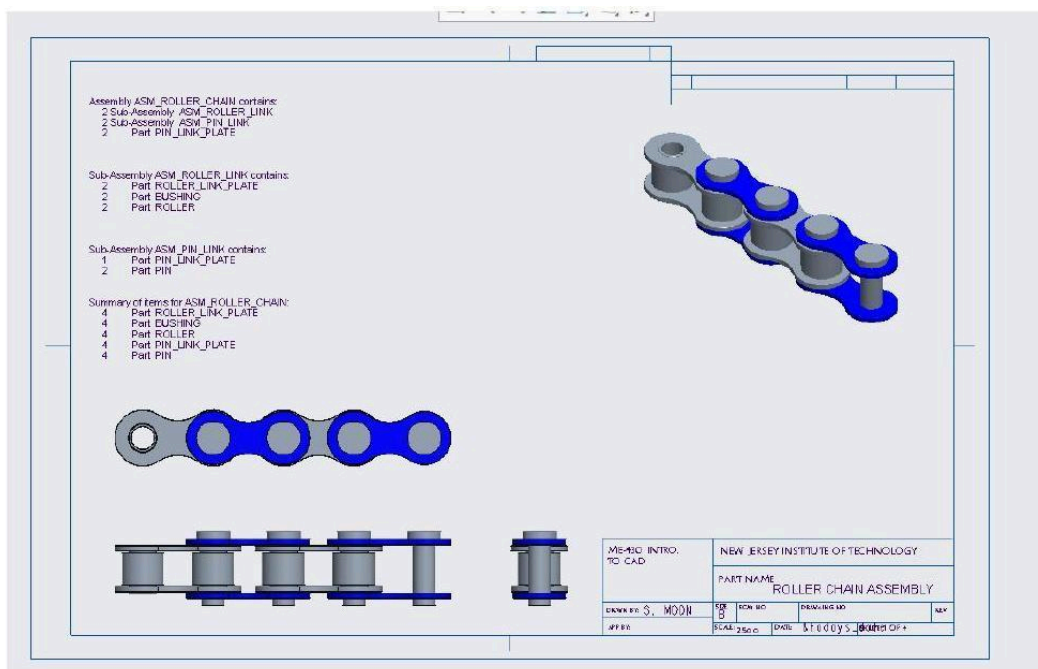


Figure 1. Roller chain assembly (top-level BOM structure, isometric, and orthographic views).

2. Design Objectives

- Model each chain component (roller link plate, pin link plate, bushing, roller, pin) as an independent parametric part, then combine them into sub-assemblies and a top-level assembly using a bottom-up modeling strategy.

- Establish a consistent chain pitch and bore/clearance scheme so that every roller link and pin link mates correctly and articulates without binding.
- Verify, through interference and clearance checking, that the assembled chain is free of part-on-part collisions at the nominal (undeflected) position.
- Produce exploded-view assemblies for both sub-modules to communicate build sequence and part relationships.
- Generate a complete, dimensioned drawing package — including bills of material, balloons, and title blocks — suitable for shop-floor fabrication and assembly.

3. Modeling Methodology

All modeling was performed in Creo Parametric using a bottom-up assembly approach. Each of the five unique components was created as a standalone part file with fully parametric, dimension-driven features (extrudes, revolves, and edge fillets), so that a change to a single controlling dimension — such as the plate bore spacing or the pin diameter — updates every dependent feature and every instance of that part in the assembly.

The parts were then grouped into two logical sub-assemblies that mirror how a physical roller chain is actually built:

- Roller Link Assembly — two roller link plates, two bushings, and two rollers, representing the outer link of the chain.
- Pin Link Assembly — one pin link plate and two pins, representing the inner link that threads through the bushings of the adjoining roller links.

The top-level ROLLER_CHAIN assembly places two instances of each sub-assembly along with two additional pin link plates, using assembly-level mates (coincident axes and mating faces) to constrain the pins concentric with the bushings and the plates flush against one another. This mirrors the physical constraint scheme of a real chain, where the pin is a slip/interference fit in the outer plate and a rotating clearance fit inside the bushing.

4. Bill of Materials and Assembly Structure

The completed chain segment contains twenty total part instances built from five unique components, summarized below. The bill of materials was generated directly from the Creo model tree, ensuring that the reported quantities always match the solid model rather than being tracked separately by hand.

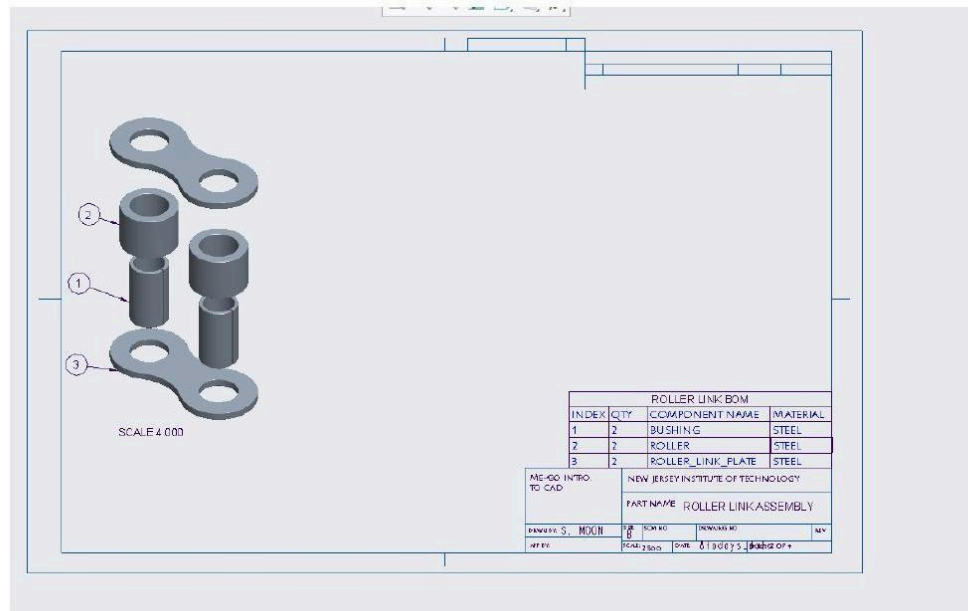


Figure 2. Roller link sub-assembly with associated bill of materials (bushing, roller, roller link plate).

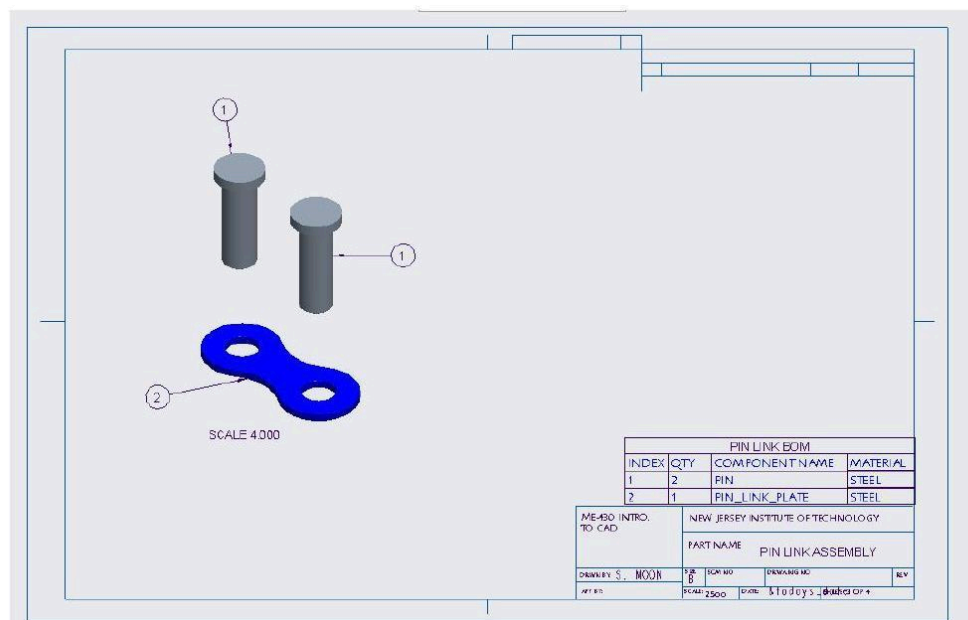


Figure 3. Pin link sub-assembly with associated bill of materials (pin, pin link plate).

5. Component Design and Key Dimensions

Dimensions were selected to give the pin a controlled clearance fit inside the bushing bore (permitting free articulation of the joint) while holding the pin link plate bore as a tight/press fit onto the pin so the two rotate together as a rigid inner link. The roller link plates carry the larger bushing/roller bore on a fixed pitch that sets the overall chain length. Small edge fillets were added to every plate to remove sharp corners that would otherwise act as stress risers under cyclic tension loading. The key values driving the model are summarized in the table below and are dimensioned in full on the pin link plate detail drawing (Figure 4).

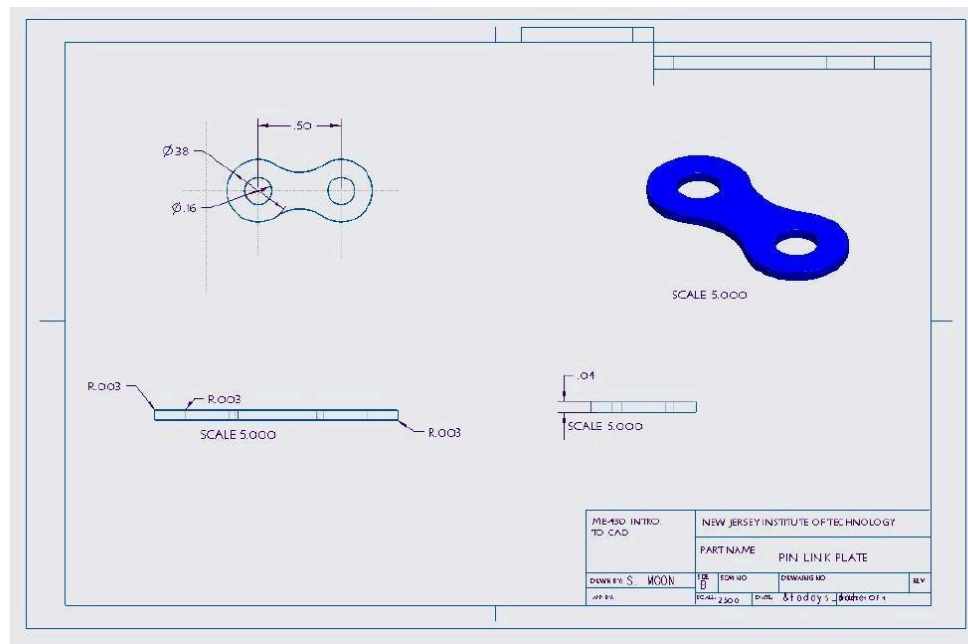


Figure 4. Pin link plate detail drawing showing bore diameters, bore spacing, plate thickness, and fillet callouts.

6. Interference Checking and Fit Verification

Because a roller chain only functions correctly if every rotating joint is free to articulate, the fully constrained assembly was run through Creo's Global Interference analysis (Analysis → Clearance and Creepage Analysis) with an exact, parts-only compute setting. This check evaluates every part-to-part boundary in the model and flags any pair of solids whose geometry overlaps at the assembled position — effectively an automated tolerance and fit check performed directly on the solid model rather than on 2D drawings.

The analysis, shown in Figure 5, returned zero interfering parts across the full twenty-part chain segment. This confirms that the pin, bushing, roller, and plate bore diameters chosen in Section 5 are dimensionally compatible at their nominal values: the pin passes through the bushing bore with running clearance, the bushing outer diameter and roller bore are concentric without collision, and the link plates stack against one another without overlap. Running this check on the assembled model — rather than only on the individual part prints — is what actually catches a tolerance stack-up problem, since two parts can each be dimensioned correctly in isolation and still interfere once assembled.

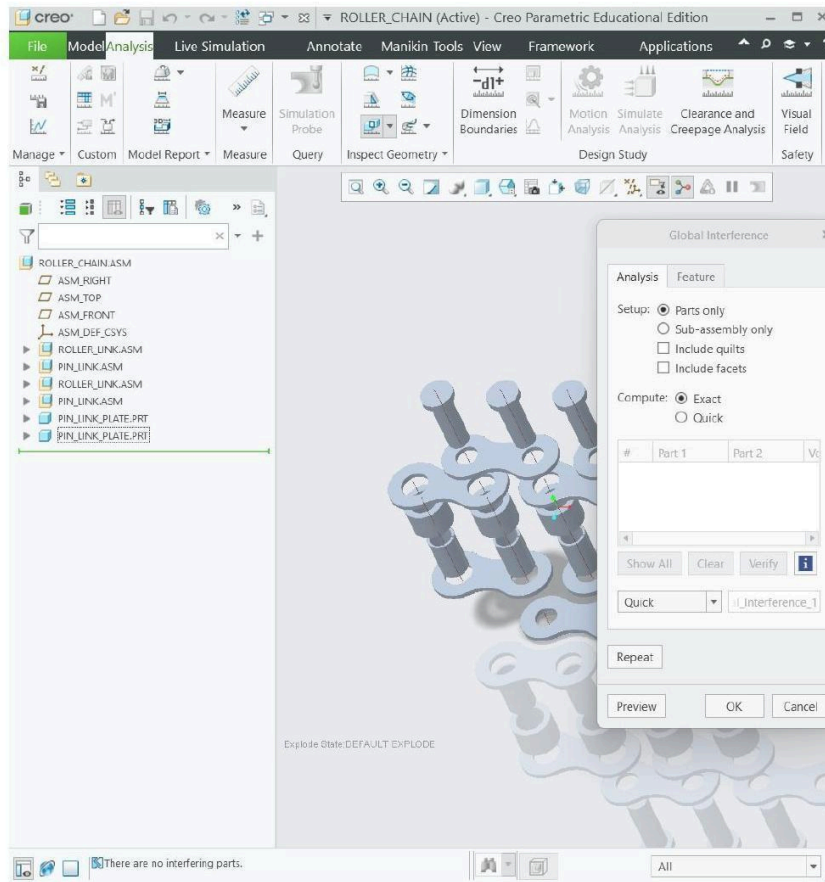


Figure 5. Global Interference analysis on the exploded ROLLER_CHAIN assembly — result: no interfering parts.

7. Exploded Views and Build Documentation

To support both assembly instructions and clear component identification, exploded views were created for each sub-assembly using Creo's Explode Position tool, with component translations driven off shared reference edges (for example, the bushing chamfer edge) so the explode direction stays consistent if the model is later modified. Item balloons were then linked to each exploded component and tied to the bill-of-materials table using quantity-split circle balloons, so the index number, quantity, and part callout stay synchronized automatically if the BOM changes.

This exploded, balloon-annotated format is standard practice for assembly drawings because it separates “what goes where” (the exploded view and balloons) from “how many and what material” (the BOM table), letting an assembler or buyer read either piece of information independently.

8. Drawing Package Deliverables

The final documentation package was built as a single multi-sheet drawing file so that the top-level assembly and every sub-assembly and detail part stay traceable to one drawing number. The package consists of:

- Sheet 1 — Roller Chain Assembly: shaded isometric and orthographic (front/top/right) views with overall length and pitch radius dimensions, plus a bill-of-materials note generated from the model tree.
- Sheet 2 — Roller Link Assembly: shaded exploded isometric view with item balloons and bill of materials.
- Sheet 3 — Pin Link Assembly: shaded exploded isometric view with item balloons and bill of materials.

- Sheet 4 — Pin Link Plate: fully dimensioned detail drawing (front, top, right-side, with hidden lines and centerlines) plus a shaded isometric view, suitable for standalone fabrication.

All sheets share a common B-size title block, decimal precision, and text style so the package reads as a single coherent set rather than four unrelated exports.

9. Results and Conclusion

The completed model demonstrates a fully parametric, twenty-part roller chain segment built from five unique components and two reusable sub-assemblies. Interference analysis confirmed an interference-free fit across every rotating and stacked joint in the assembly, validating the bore, pin, and bushing dimensions chosen during part design. The resulting drawing package — assembly views, exploded sub-assembly views with balloons, and a fully dimensioned detail drawing — provides everything needed to fabricate the individual parts and to assemble them correctly, and mirrors the documentation an engineering team would hand off to a machine shop for a production run.

Beyond the specific chain geometry, the project reinforced a bottom-up assembly workflow — parametric parts → logical sub-assemblies → top-level assembly — together with model-driven verification (interference/clearance checking) as a routine step before a design is released for drawings, rather than as an afterthought performed only on paper.